

COMICCRAFT – AI COMIC STORY CREATOR

Project Planning Phase

Project Type: Web-based AI Comic Story Generator

Primary Technologies: FastAPI, Google Gemini Models, Stable Diffusion, Hugging Face Diffusers, Jinja2, FPDF, PyTorch

Planning Basis: Project specification and implementation workflow in the uploaded ComicCraft document.

1. Planning Overview

The planning phase defines how ComicCraft will be developed as an end-to-end web application that accepts a user's comic idea and preferences, generates a structured story, produces illustrations, presents the result as comic panels, and exports the completed comic as a PDF.

The project plan is organized around model selection, system architecture, environment setup, core functionality, frontend development, routing, testing-oriented checks, and local deployment.

2. Project Objectives

- Create a web application for generating personalized comic stories from user prompts.
- Accept story prompt, character name, setting, tone, and art style as user inputs.
- Generate a structured five-panel comic outline using Gemini Flash.
- Generate narration and dialogue using Gemini Pro.
- Generate comic-style illustrations using Stable Diffusion through Hugging Face Diffusers.
- Combine generated text and images into a panel-based comic layout.
- Provide a browser-based preview of the generated comic.
- Export the completed comic into a downloadable multi-page PDF.

3. Planned System Architecture

Layer	Planned Component	Main Responsibility
Frontend	HTML / CSS / Jinja2	Collect inputs and display comic preview/results
Backend	FastAPI	Handle requests, workflow orchestration and routes
Story AI	Gemini Flash	Create structured five-panel outline
Narration AI	Gemini Pro	Create narration and dialogue
Image AI	Stable Diffusion + Diffusers	Generate comic illustrations
Layout	layout_builder.py	Match story content with generated images
Export	FPDF / exporters.py	Create the final PDF comic

4. Development Work Plan

Phase 1 – Model Selection and Architecture

Define the AI requirements and select Gemini Flash for structured outlines, Gemini Pro for detailed narration/dialogue, and Stable Diffusion for illustrations. Establish the frontend, backend, AI integration, layout and export architecture.

Phase 2 – Environment Setup

Prepare Python, pip and a virtual environment; install the required FastAPI, Uvicorn, Google Gemini, Hugging Face Diffusers, PyTorch and FPDF dependencies; configure API keys through environment variables.

Phase 3 – Core AI Functionality

Implement outline generation, story generation and image generation. Organize the generated outputs into a consistent panel structure.

Phase 4 – Backend and Routing

Implement FastAPI routes including the homepage, comic generation, JSON generation, export-success page and direct image testing route. Connect the routes to the AI workflow.

Phase 5 – Frontend Development

Build the input page, comic preview page and export-success page using HTML, CSS and Jinja2 templates.

Phase 6 – Layout and PDF Export

Match panel images with titles, scene descriptions, captions and narration, then create the multi-page PDF using FPDF.

Phase 7 – Local Deployment and Verification

Run the application with Uvicorn, verify the main browser flow and API documentation endpoint, and check the generated comic and PDF output.

5. Planned User Workflow

- User opens the ComicCraft homepage.
- User enters a story prompt and optional character, setting, tone and art-style preferences.
- The backend receives the form request.
- Gemini Flash creates a five-panel outline containing panel information and image prompts.
- Gemini Pro generates narration and dialogue for the story.
- Stable Diffusion generates illustrations for the panels.
- The application organizes the generated content into a comic layout.
- The comic preview page displays panel titles, images, scene descriptions and captions/narration.
- The user downloads the completed comic as a PDF.
- The export-success page confirms completion and provides an option to create another comic.

6. Planned Project Structure

Component	Planning Role
routes.py	Central FastAPI request routing and AI workflow coordination
gemini_flash.py	Five-panel outline generation
gemini_pro.py	Narration and dialogue generation
image_generator.py	Stable Diffusion image generation and storage
layout_builder.py	Assembly of story information and panel images
exporters.py	PDF creation using FPDF
templates/index.html	User input interface
templates/comic_preview.html	Generated comic preview
templates/export_success.html	Export confirmation page
static/panels	Generated panel images
static/exports	Generated PDF exports

7. Planned Resources and Prerequisites

- Python and pip for the application environment.
- FastAPI and Uvicorn for the web backend and local server.
- HTML, CSS and Jinja2 for the web interface and dynamic templates.
- Google Gemini API access for story generation.
- Hugging Face Diffusers and Stable Diffusion for image generation.
- PyTorch as the supporting machine-learning framework.
- FPDF for PDF export.
- Git and virtualenv for project management and environment isolation.

8. Planned Configuration

The project document specifies API keys through environment variables, including **GEMINI_API_KEY** and **HF_API_KEY**. The application is planned to run locally with Uvicorn and be accessible through the local FastAPI server, with the API documentation available through the /docs endpoint.

9. Milestones and Deliverables

Milestone	Expected Deliverable
Requirements and planning	Defined ComicCraft workflow and technology requirements
Architecture	Frontend, FastAPI backend and AI integration structure
AI integration	Outline, narration/dialogue and image generation functions
Backend	FastAPI routes connected to the generation pipeline
Frontend	Input, preview and export-success templates
Comic assembly	Panel layout combining generated text and images
PDF export	Downloadable multi-page comic PDF
Deployment check	Locally running ComicCraft application and API documentation

10. Planning Checklist

- Define user inputs and comic-generation workflow.
- Select and integrate the planned AI models.
- Prepare Python virtual environment and dependencies.
- Configure required API keys.
- Implement story outline, narration/dialogue and image generation.
- Connect AI functions through FastAPI routes.
- Develop Jinja2-based frontend templates.
- Build the comic panel layout.
- Implement PDF export.
- Run the application locally and verify the complete workflow.

11. Planning Outcome

The planning phase establishes a clear development path from user input to AI-generated comic output. The planned workflow separates responsibilities across the FastAPI backend, Gemini models, Stable Diffusion image generation, Jinja2 frontend templates, comic layout assembly and FPDF export. This structure provides the foundation for implementing the ComicCraft application as described in the project document.

Source basis: ComicCraft – AI Comic Story Creator using Gemini Models project document.